Student solves scientific puzzle that could improve solar panel efficiency and increase usage

A Loughborough University PhD student, Tom Fiducia, has helped shed light on a solar panel puzzle that could lead to more efficient devices being developed. Most of the world's solar power is currently produced by solar panels—also known as photovoltaic panels—that are made of silicon.

New solar panels have been created that are made from a semiconducting material called cadmium telluride (CdTe). CdTe panels have been found to produce electricity at lower costs than silicon panels, and there has been a dramatic gain in efficiency brought about by adding an element called selenium to CdTe. As a result, electricity from CdTe solar farms is being produced more cheaply than it is from fossil fuels, giving economic as well as environmental benefits around the world.

Until now, it was not well-understood why selenium increases efficiency, but thanks to Fiducia and an international team of researchers, the puzzle has been solved. Fiducia has worked with leading solar experts from Centre for Renewable Energy Systems Technology (CREST), Durham University, University of Oxford and Colorado State University to explore the effect selenium has on solar panels.

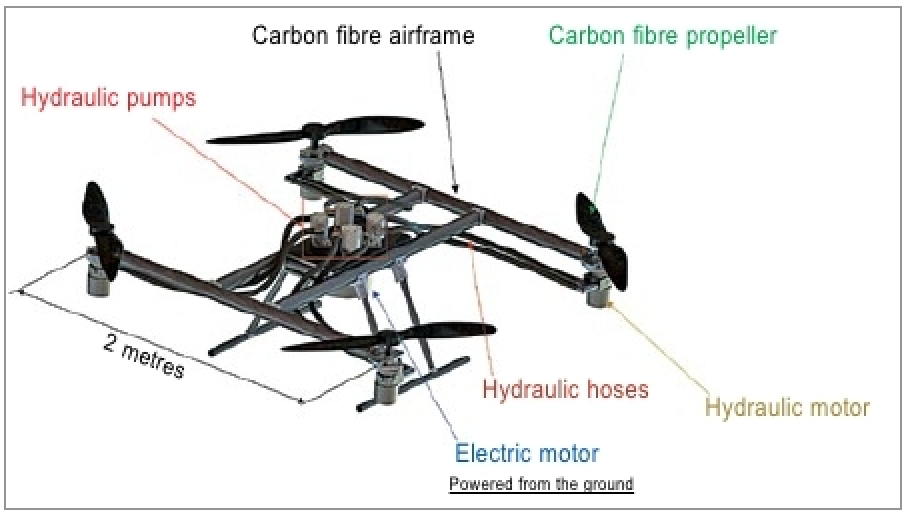
Their research has revealed that selenium works by overcoming the effect of harmful, atomic-scale defects in CdTe panels. This explains the increase of efficiency as electrons, which are generated when sunlight hits the solar panel, are less likely to be trapped and lost at the defects. This increases the amount of power extracted.

Advancement in driverless aircraft could direct the future of drones, flight

Billions of dollars are being spent by aviation giants and aerospace startups to create driverless flying vehicles that can meet the growing need for rapid and flexible travel and delivery. One of the solutions currently being worked on is the autonomous aerial vehicle (AAV) with vertical takeoff and landing (VTOL) ability. The technology is currently being tested to carry cargo and eventually taxi passengers.

Lizhi Shang, a postdoctoral research assistant who works on the technology with Andrea Vacca, professor of agricultural and biological engineering at Purdue University, says, "Everyone is facing the same problem with weight in creating these types of vehicles. Drones require heavy batteries and lots of electrical components, which leaves little room for the actual payload."

Shang also says that many current systems are expensive,



Researchers at Purdue University have come up with a method to use fluid power technology for the future of drones and flight (Credit: Purdue University)